System Rules (Instructions)

Task: Generate a data science pipeline in Python 3.10.

Input: A dataset in CSV format, a schema that describes the columns and data types of the dataset, and a data profiling info that summarizes the statistics and quality of the dataset.

Output: A pipeline code that performs the following steps:

#1: Import the necessary libraries and modules.

#2: Load the training and test datasets. For the training data, utilize the variable "train_data.csv", and for the test data, employ the variable "test_data.csv".

#3: The user will provide the Schema, and Data Profiling Info of the dataset with columns appropriately named as attributes.

#4: Perform missing value imputation for features 'Address' and 'Zip'.

#5: Perform feature extraction (dataset contains categorical values).

#6: Perform feature selection (redundant columns e.g., 'Address' and 'Zip').

#7: Select an appropriate ML algorithm.

#8: Assign a default value to a particular hyperparameter.

#9: Evaluate the model.

Catalog Data

Dataset Description: A The dataset was obtained from multiple sources, including surveys, job posting sites, and other publicly available sources.A total of 6704 data points were collected.The dataset included fivevariables: age, experience, job role, and education level and salary.

Schema, and Data Profiling Info:

Experience (string), distinct [60%], missing [0%], categorical-vals [1 year, 2 years, 3 years]
Gender (string), distinct [40%], missing [0%], categorical-vals [Male, Female]
Address (string), distinct [40%], missing [20%], categorical-vals [CA, TX]
Zip (string), distinct [40%], missing [40%], categorical-vals [7050, 7871]
Salary (int, **target feature**), min-max vals [100, 310], total-vals [5]

Categorical Features: Experience, Gender, Address, Zip